unfolding of cosmic history, and, surely, is one of humanity's most profound discoveries, let's briefly examine how we know it is so.

In 1929, Edwin Hubble, using the 100-inch telescope at the Mount Wilson observatory in Pasadena, California, found that the couple of dozen galaxies he could detect were all rushing away.[5] In fact, Hubble found that the more distant a galaxy is, the faster its recession. To give a sense of scale, more refined versions of Hubble's original observations (that have studied thousands of galaxies using, among other equipment, the Hubble Space Telescope) show that galaxies that are 100 million light-years from us are moving away at about 5.5 million miles per hour, those at 200 million light-years are moving away twice as fast, at about 11 million miles per hour, those at 300 million light-years' distance are moving away three times as fast, at about 16.5 million miles per hour, and so on. Hubble's was a shocking discovery because the prevailing scientific and philosophical prejudice held that the universe was, on its largest scales, static, eternal, fixed, and unchanging. But in one stroke, Hubble shattered that view. And in a wonderful confluence of experiment and theory, Einstein's general relativity was able to provide a beautiful explanation for Hubble's discovery.

Actually, you might not think that coming up with an explanation would be particularly difficult. After all, if you were to pass by a factory and see all sorts of material violently flying outward in all directions, you would likely think that there had been an explosion. And if you traveled backward along the paths taken by the scraps of metal and chunks of concrete, you'd find them all converging on a location that would be a likely contender for where the explosion occurred. By the same reasoning, since the view from earth—as attested to by Hubble's and subsequent observations—shows that galaxies are rushing outward, you might think our position in space was the location of an ancient explosion that uniformly spewed out the raw material of stars and galaxies. The problem with this theory, though, is that it singles out one region of space—our region—as unique by making it the universe's birthplace. And were that the case, it would entail a deep-seated asymmetry: the physical conditions in regions far from the primordial explosion—far from us—would be very different from those here. As there is no evidence for such asymmetry in astronomical data, and furthermore, as we are highly suspect of anthropocentric explanations